

# Lost in the Middle: Context Retrieval Accuracy in Large Language Models

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## Abstract

Large language models are increasingly used to extract structured information from long documents, but prior work suggests that they may not use all parts of a context equally well. This project studies whether structured extraction accuracy depends on where relevant information appears within a document. An initial version of the experiment produced degenerate results, with perfect accuracy across all conditions, so the document generator was redesigned to embed target facts into longer and more realistic text with additional distractors. In the revised experiment, exact-match accuracy remains perfect when target information appears near the beginning or end of the document, but drops slightly when it appears in the middle, providing small but suggestive evidence of a positional retrieval effect.

## 1 Introduction

Large language models are now routinely used to work with long documents. In practice, a user may provide a policy, contract, internal report, or planning memo and ask the model to identify a specific fact or extract a small number of structured fields. In these settings, the relevant information may appear anywhere in the document and is often surrounded by irrelevant or distracting text.

This project asks a simple question with practical importance: does the location of relevant information inside a document affect how accurately a language model retrieves it? More specifically, we study whether structured extraction becomes less reliable when the target information appears in the middle of a document rather than near its beginning or end.

Recent work on long-context behavior in transformer models suggests that retrieval quality may not be uniform across positions. The theoretical foundation for modern LLMs is the transformer architecture introduced by Vaswani et al. [1]. Transformers rely on self-attention, which allows tokens to interact across the sequence and makes long-context processing possible. However, the ability to attend across a sequence does not imply that all positions are used equally well. Liu et al. [3] show that many language models perform best when relevant information appears near the beginning or end of a long input and worse when the same information appears in the middle. Their work directly motivates the central question of this project.

Other work has examined the broader difficulty of generalizing to long contexts. Press, Smith, and Levy [4] show that positional representations and attention mechanisms affect how models extrapolate to longer inputs. The GPT-4 technical report also notes that long-context reasoning remains challenging even when a model supports large context windows [5]. Brown et al. [2] demonstrate the broad capabilities of large language models, but later work has made clear that strong general capability does not always imply consistent retrieval reliability in long-context settings.

Motivated by these findings, this project studies positional effects in a controlled structured extraction task. Rather than using an existing benchmark, we construct a synthetic but carefully designed dataset in which the same type of information can appear at the beginning, middle, or end of a document. The goal is to isolate whether retrieval accuracy depends on position when the overall task remains the same.

A central part of the project is methodological. The first version of the experiment proved too easy and produced perfect accuracy across all conditions. Because of this, the document generator was redesigned to embed target facts inside longer and more realistic text with additional distractors. The final experiment therefore focuses not only on the results, but also on how the redesign of the data generation process made it possible to observe meaningful variation in model behavior.

## 2 Methods

Our task is a structured extraction problem. Each document describes multiple fictional employees, and the model is asked to extract information about one target employee. The required fields

are `job_title`, `department`, `years_at_company`, and `weekly_hours`. The model is instructed to return valid JSON with exactly these keys and to rely only on information explicitly stated in the document.

The project began with a simpler document generator that produced short synthetic company documents and used a fixed target employee. In that early version, the four target attributes usually appeared together in one clearly identifiable paragraph. The result was a degenerate experiment: the model achieved perfect accuracy in the begin, middle, and end conditions. That outcome showed that the pipeline was operational, but it also showed that the task was too easy to test the research question in a meaningful way.

Because the initial experiment produced degenerate results, the document generator was redesigned to create a more meaningful and realistic evaluation setting. The goal of the redesign was to make the extraction task harder and to ensure that correct retrieval required the model to process the full document rather than rely on obvious patterns.

The revised generator randomizes the target employee name, distributes target facts across multiple sentences, introduces additional distractor employees with overlapping or semantically similar attributes, and embeds the relevant information within longer sections such as company overview, planning notes, staffing updates, and project summaries. Instead of presenting the answer in a single isolated paragraph, the target information is now embedded within larger and more realistic text chunks, making the retrieval task closer to real document-processing scenarios. The revised generator randomizes the target employee name, distributes target facts across multiple sentences, introduces more distractor employees with overlapping or semantically similar attributes, and embeds the relevant information within longer sections such as company overview, planning notes, staffing updates, and project summaries. The key idea was to stop presenting the answer as an isolated paragraph and instead embed it within larger and more realistic text chunks.

The revised dataset contains 300 synthetic documents in total, with 100 documents in each positional condition. In the begin condition, the target information appears near the beginning of the document. In the middle condition, it appears near the middle. In the end condition, it appears near the end. Ground-truth labels are stored in a JSON Lines file keyed by document identifier so that model outputs can be matched back to the correct labels automatically during evaluation.

The extraction prompt asks the model to identify information about a named target employee

and return a JSON object. It explicitly instructs the model to use only stated information, ignore other employees, avoid guessing, and return valid JSON with no extra text. The model used in the final experiment is `gpt-4o-mini`, run with temperature set to zero in order to reduce unnecessary generation variability and make the positional comparison easier to interpret.

The full experimental workflow is straightforward. We first generate documents with controlled placement of the target facts, then fill a prompt template with the document text and target employee name, then query the model, then parse the returned JSON, and finally compare the prediction with the ground truth. Because the generator, prompt, extraction script, and evaluation script are separated in the repository, it was possible to revise the document generator without changing the scoring framework.

We evaluate the outputs in three ways. The main metric is exact-match accuracy, which counts a document as correct only when all four fields match the ground truth. We also compute field-level accuracy so that partial failures can be distinguished from complete failures. Finally, we record JSON validity, since a downstream system would fail if the returned format were invalid even when the content was approximately correct.

### 3 Results

The revised experiment yields a small but informative positional pattern. Table 1 reports exact-match accuracy by document position. The observed positional pattern is consistent with the hypothesis that retrieval from long contexts is not uniform across the sequence. Even though the model can attend to the entire document, information located near the middle of the context may compete with surrounding text, especially when multiple employees with similar attributes appear in nearby sections. This competition may reduce the likelihood that the model correctly associates the target name with the correct attribute values.

| Position | Exact-Match Accuracy |
|----------|----------------------|
| Begin    | 1.00                 |
| Middle   | 0.97                 |
| End      | 1.00                 |

Table 1: Exact-match accuracy by context position.

Exact-match accuracy is perfect when the target information appears near the beginning or the end of the document, but drops slightly when the information appears in the middle. The magnitude of the difference is small, but it is consistent across the dataset. Because the task requires matching multiple attributes to the same employee name, even a small amount of confusion can cause the entire prediction to be counted as incorrect under exact-match evaluation. This makes exact-match accuracy a strict metric that can reveal subtle retrieval failures that would not appear in a looser evaluation. The absolute difference is modest, but the direction of the effect is consistent with the *lost in the middle* hypothesis and is much more informative than the original degenerate setup in which all conditions achieved the same perfect score.

Field-level results show that the observed errors are concentrated in one attribute. As Table 2 shows, the reduction appears only in the department field. This pattern suggests that positional effects do not affect all attributes equally. Department labels are short categorical phrases that may overlap semantically with other departments mentioned elsewhere in the document, making them more vulnerable to interference. In contrast, years at company and weekly hours are numeric values that tend to be unique, and job titles are often strongly associated with a specific employee name, which may make them easier for the model to retrieve even in the middle of the context.

| Field            | Begin | Middle | End  |
|------------------|-------|--------|------|
| Job Title        | 1.00  | 1.00   | 1.00 |
| Department       | 1.00  | 0.97   | 1.00 |
| Years at Company | 1.00  | 1.00   | 1.00 |
| Weekly Hours     | 1.00  | 1.00   | 1.00 |

Table 2: Field-level accuracy by context position.

JSON validity remains 1.00 in all three conditions. This indicates that the model reliably follows the output format instructions even when it makes a retrieval mistake. The failures in the revised experiment are therefore content-level retrieval errors rather than formatting errors.

Figure 1 visualizes exact-match accuracy by position, and Figure 2 shows the field-level breakdown. Together they make the main empirical pattern clear: the middle condition is slightly less reliable, and the difference is concentrated in department extraction rather than spread uniformly across all fields.

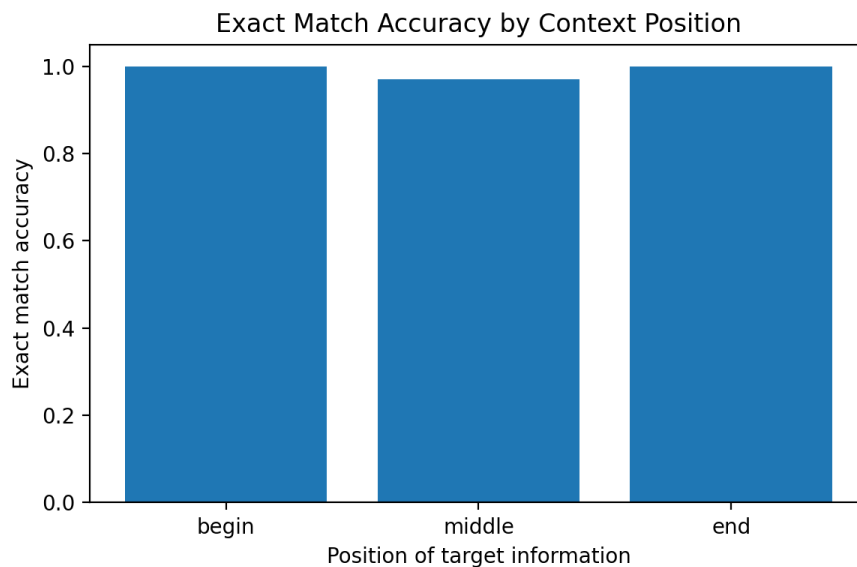


Figure 1: Exact-match accuracy by context position. The middle condition shows a small drop relative to the begin and end conditions.

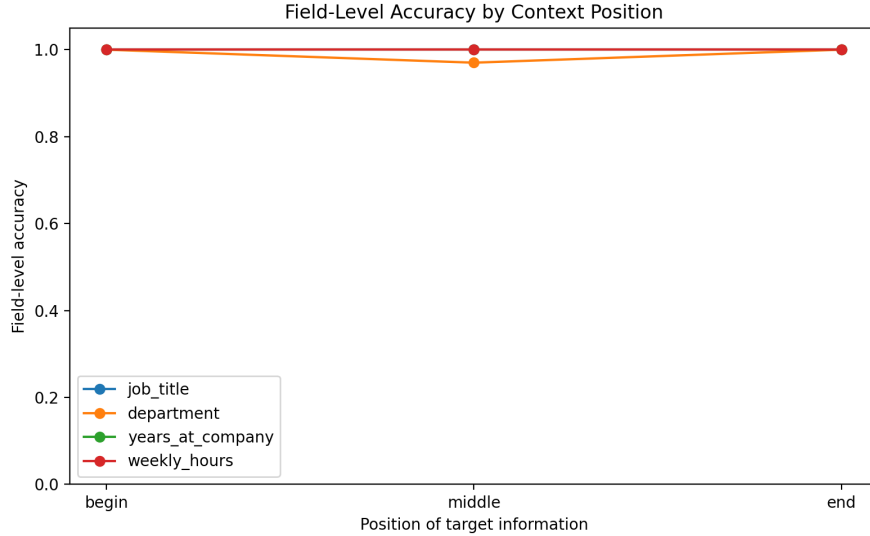


Figure 2: Field-level accuracy by context position. The small accuracy reduction is concentrated in the department field.

The size of the effect should be interpreted carefully. With 100 documents per condition, the reduction from 1.00 to 0.97 corresponds to only three errors in the middle condition. A simple comparison of proportions does not indicate a statistically significant difference at this sample size, so the observed gap should not be interpreted as strong evidence of a positional failure.

However, the direction of the effect is consistent across the dataset and aligns with the pattern reported in prior work on long-context retrieval. Rather than showing a large performance drop, the results suggest that a small disadvantage for middle-position information may become visible once the task is made more realistic and contains more distractors. For this reason, the results should be viewed as suggestive evidence of a positional retrieval effect rather than a definitive measurement of its magnitude.

For this reason, the results should be viewed as suggestive evidence of a positional retrieval effect rather than a definitive measurement of its magnitude. A larger dataset or repeated runs with different random seeds would be needed to determine whether the effect is statistically reliable.

Another possible explanation for the middle-position errors is attention dilution. When the document becomes longer and contains several employees with similar roles, the model must distribute attention across many tokens. Information located near the beginning or end may receive stronger effective attention, while information in the middle may be surrounded by competing

signals, making correct extraction slightly less reliable.

## 4 Discussion

The main result of the project is not simply that the middle condition performs slightly worse. This outcome highlights the importance of experimental design when studying long-context behavior. If the task is too simple, the model may achieve perfect accuracy regardless of where the relevant information appears, which can hide meaningful differences between positions. By embedding the target facts inside longer and more realistic text, the revised generator created a setting in which retrieval errors could occur, making it possible to observe positional effects that were not visible in the original experiment. Equally important is the fact that this pattern became visible only after the experimental design was improved. The first version of the dataset was too simple: the target employee was fixed, the relevant facts were grouped together, and the answer could often be recovered by locating one obvious paragraph. In that setting, the model’s perfect accuracy said more about the weakness of the task than about the absence of positional effects.

Once the generator was redesigned to produce longer documents with more distractors and more distributed target information, a positional pattern emerged. That pattern is small, but it is aligned with prior work on long-context retrieval and therefore strengthens the interpretation that document position can matter even in relatively simple extraction tasks. Prior work has shown that transformer-based models may attend more strongly to tokens near the beginning or end of the sequence, while information located in the middle can be harder to recover when the context becomes long. Although our experiment is much simpler than the benchmarks used in previous studies, the direction of the effect is consistent with the lost-in-the-middle phenomenon reported in the literature. This suggests that even in structured extraction tasks, positional bias may still influence model behavior.

The field-level results also suggest that positional effects do not necessarily appear as a uniform decline across all outputs. In our experiment, the only field that became less reliable in the middle condition was **department**. This is plausible because department labels are short categorical names that may overlap semantically with other labels appearing elsewhere in the same document. By contrast, weekly hours and years at company are numeric and relatively distinctive, while job titles



may be strongly associated with the target employee’s name. The observed errors therefore look less like random noise and more like confusions caused by contextual competition.

This interpretation is relevant for practice. Real documents are often longer, noisier, and more repetitive than the synthetic documents used here. They may contain many entities, repeated concepts, inconsistent formatting, and several semantically similar labels. If a small middle-position disadvantage already appears in this controlled experiment, it is reasonable to suspect that more realistic production settings could amplify the problem. That does not mean long-context models are ineffective; rather, it suggests that document structure, passage selection, redundancy checks, and field-specific validation may still matter even when large context windows are available.

## 5 Limitations and Future Work

This study has several limitations. The dataset is still small, the documents are synthetic. Because the documents are generated synthetically, they may not fully reflect the complexity of real-world text. Real documents often contain inconsistent formatting, incomplete information, and multiple references to the same entity across different sections. These factors could make retrieval more difficult and might increase the size of positional effects compared to the controlled setting used in this experiment. And the task is relatively simple. Even after redesign, the task remains simpler than many real extraction problems. Each document contains only one target employee, and the required fields are limited to four attributes. More challenging settings could include multiple target entities, partially conflicting information, or attributes that must be inferred from several sentences. Such conditions might produce a larger middle-position disadvantage. And only one model configuration is evaluated. The effect size is also modest. These results should therefore be interpreted as suggestive rather than definitive. They show that a positional effect can emerge under a more realistic generator design, but they do not establish a large or universal middle-position bias across all models and all retrieval settings.

Several natural extensions would strengthen the study. A larger dataset would make the analysis more stable and would allow confidence intervals or formal significance testing. The documents themselves could be made longer and more varied, with repeated mentions of employees across sections. The experiment could also compare multiple models or prompt formulations, or introduce

harder settings such as partially conflicting information, missing fields, or multi-step reasoning requirements.

## 6 Conclusion

This project examined whether a language model’s ability to retrieve factual information depends on where that information appears within a document. Using a controlled structured extraction task, we compared begin, middle, and end placements of target facts in synthetic company-style documents.

The first version of the experiment produced degenerate results, with perfect accuracy across all positions. Rather than treating that outcome as final, we redesigned the document generator so that target facts were embedded in longer and more realistic contexts with more distractors. That redesign was essential to the project. Once the task became less trivial, a small but clear positional effect appeared.

In the revised experiment, exact-match accuracy remained perfect when the target information appeared near the beginning or end of the document, but dropped slightly when the information appeared in the middle. The resulting errors were concentrated in the department field, suggesting that middle-position information may be especially vulnerable when the attribute itself is somewhat ambiguous or easy to confuse with nearby distractors. Taken together, the results support a measured conclusion: even in a relatively simple structured extraction task, a middle-position disadvantage can emerge once the data generation process is realistic enough to reveal it.

## Acknowledgments

We thank the course staff for feedback on the project design, especially the suggestion to embed target information into larger text chunks. That feedback directly motivated the redesign that made the final experiment more informative.

## A Appendix A: Final Extraction Prompt

**Description.** Prompt template used in the final extraction experiment. The placeholder fields were filled at runtime with the target employee name and document text.

You will be given a long company document containing background information, team updates, project notes, and employee-related details.

Your task is to extract information about the employee named `{{TARGET_NAME}}`.

IMPORTANT RULES:

- Use only information explicitly stated in the document.
- The relevant facts may be spread across different parts of the document.
- Ignore information about other employees, even if their roles or departments seem similar.
- Do not guess, infer, or fill in missing values.
- If a field is not explicitly stated, return null.
- Return valid JSON only, with no extra text.

Return JSON with exactly these keys:

- `job_title`
- `department`
- `years_at_company`
- `weekly_hours`

Document:

`{{DOCUMENT}}`

## B Appendix B: Example Document Snippet

**Description.** Representative excerpt from a revised synthetic document. The full documents were longer and included multiple sections, distractor employees, and background narrative.

### Company Overview

The company has continued expanding its internal reporting processes in response to growth across finance, operations, and product functions. Leadership has emphasized documentation quality, metric consistency, and stronger cross-functional visibility in quarterly planning cycles.

### Planning Notes

According to the latest staffing summary, Mina Chen's role is listed as Data Analyst. Internal staffing notes place Mina Chen in Finance Analytics. Another planning note briefly referenced Mina Chen in the context of ongoing cross-team coordination.

### Staffing and Project Updates

Current records indicate that Mina Chen has spent 3 years at the company. Workload notes indicate that Mina Chen generally maintains a 40-hour work week.

## C Appendix C: Example Ground Truth and Model Output

**Description.** Illustrative example of one evaluation pair used in scoring.

Ground truth:

```
{  
  "job_title": "Data Analyst",  
  "department": "Finance Analytics",
```

```
"years_at_company": 3,  
"weekly_hours": 40  
}
```

Model output:

```
{  
  "job_title": "Data Analyst",  
  "department": "Finance Analytics",  
  "years_at_company": 3,  
  "weekly_hours": 40  
}
```

## D Appendix D: Example Error Case

**Description.** Example of the kind of middle-condition error observed in the revised experiment.

Ground truth:

```
{  
  "job_title": "Senior Data Analyst",  
  "department": "Marketing Analytics",  
  "years_at_company": 7,  
  "weekly_hours": 35  
}
```

Model output:

```
{  
  "job_title": "Senior Data Analyst",  
  "department": "Product Analytics",  
  "years_at_company": 7,  
  "weekly_hours": 35  
}
```

## E Appendix E: Error Count by Position

**Description.** Error count implied by the main exact-match results.

| Position | Documents | Exact-Match Errors |
|----------|-----------|--------------------|
| Begin    | 100       | 0                  |
| Middle   | 100       | 3                  |
| End      | 100       | 0                  |

Table 3: Error counts by position derived from the exact-match accuracy values reported in the main text.

## References

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